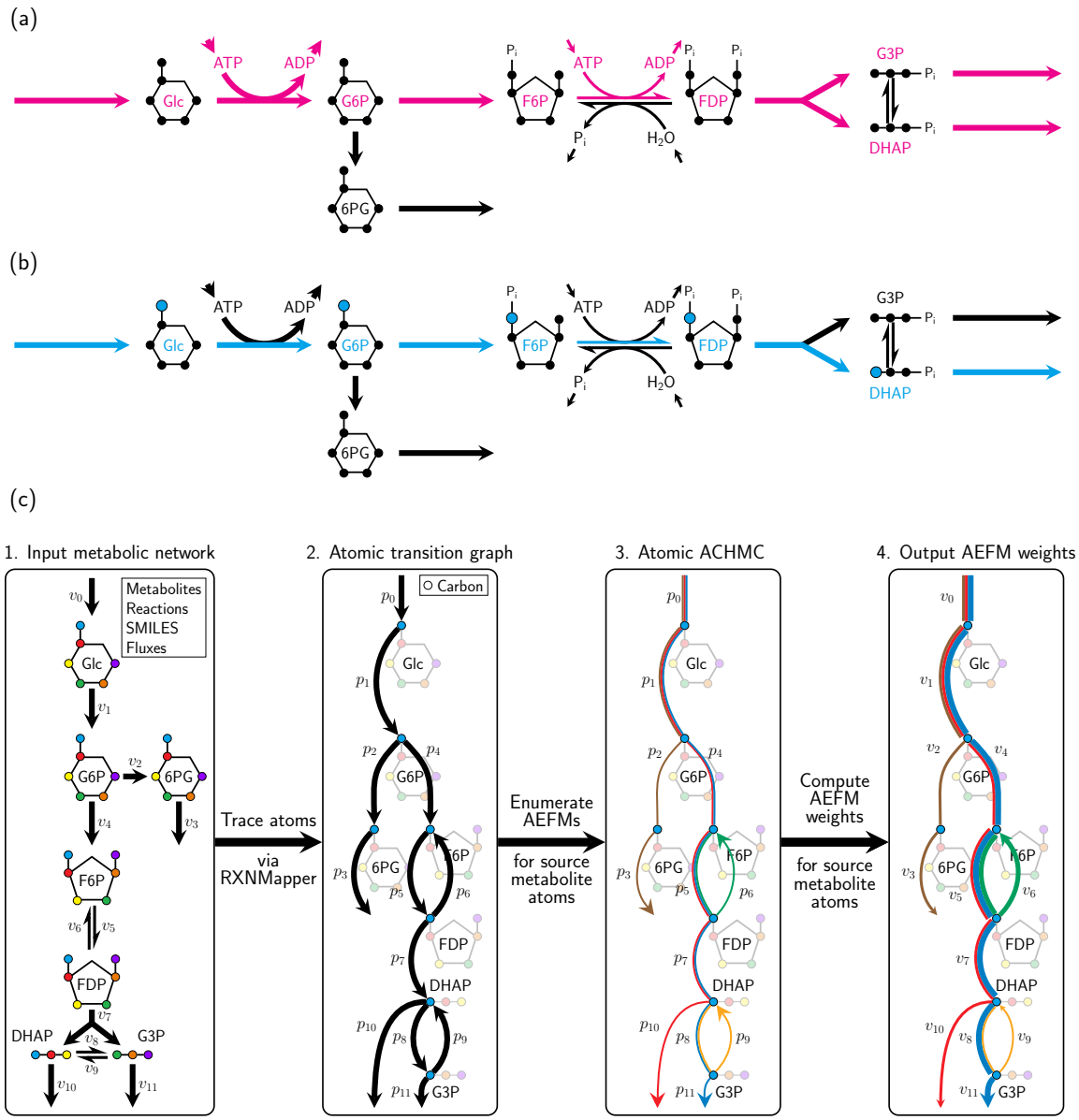

Figure 1: Atomic cycle-history Markov chain (ACHMC) pipeline. (a) An example molecular EFM in glycolysis. (b) An example AEFM tracing the flow of a given glucose carbon in glycolysis. (c) The major steps within our pipeline to enumerate AEFMs and compute their weights.

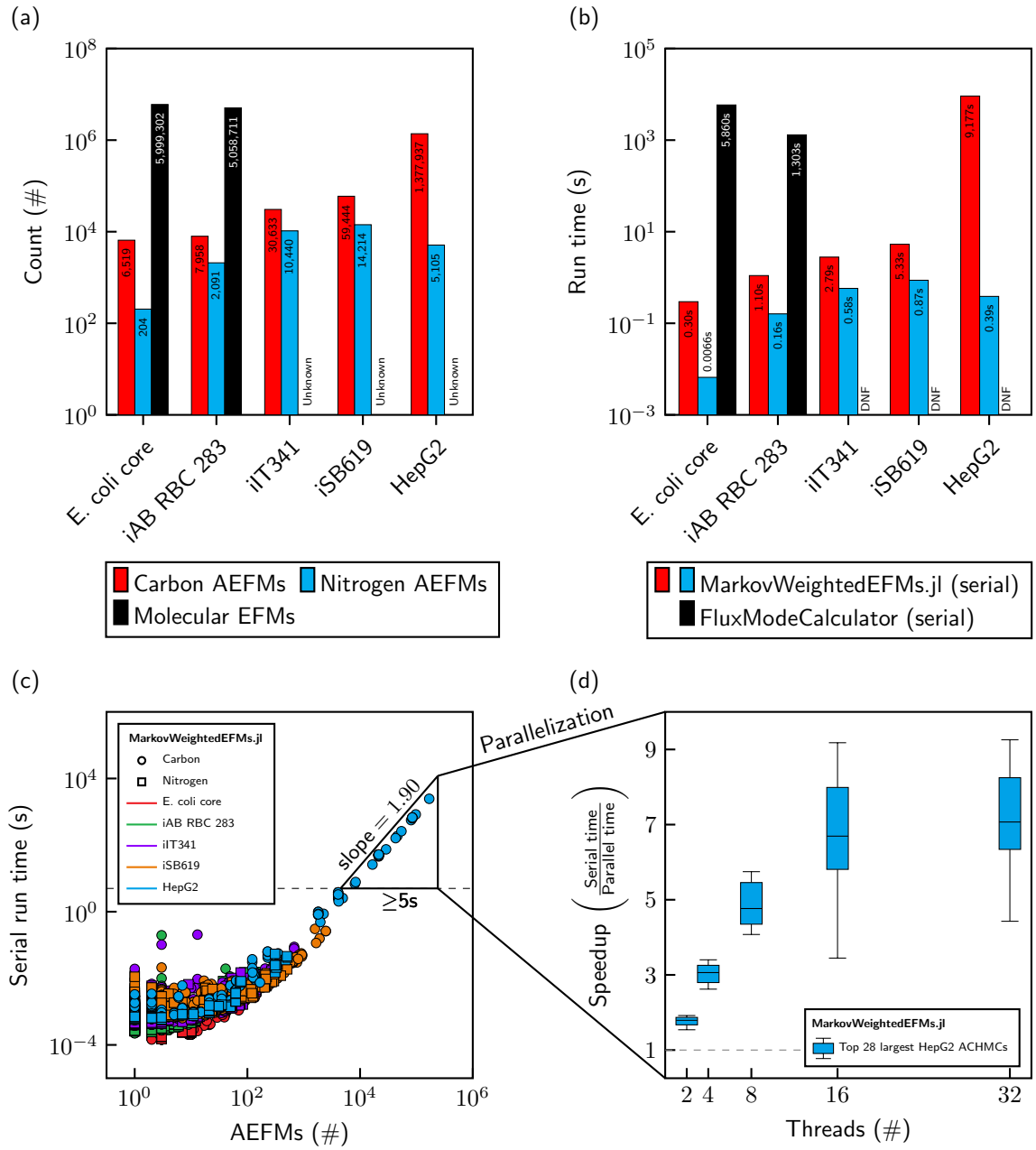

Figure 2: AEFMs are computationally tractable compared to molecular EFMs in five GEMs. (a) Number of atomic and molecular EFMs in the five GEMs. (b) The run time of our MarkovWeightedEFMs.jl method versus FluxModeCalculator running in serial. (c) Serial run time scaling of MarkovWeightedEFMs.jl as a function of the number of atomic EFMs. (d) Speedup of selected ACHMCs as a functional of additional threads.

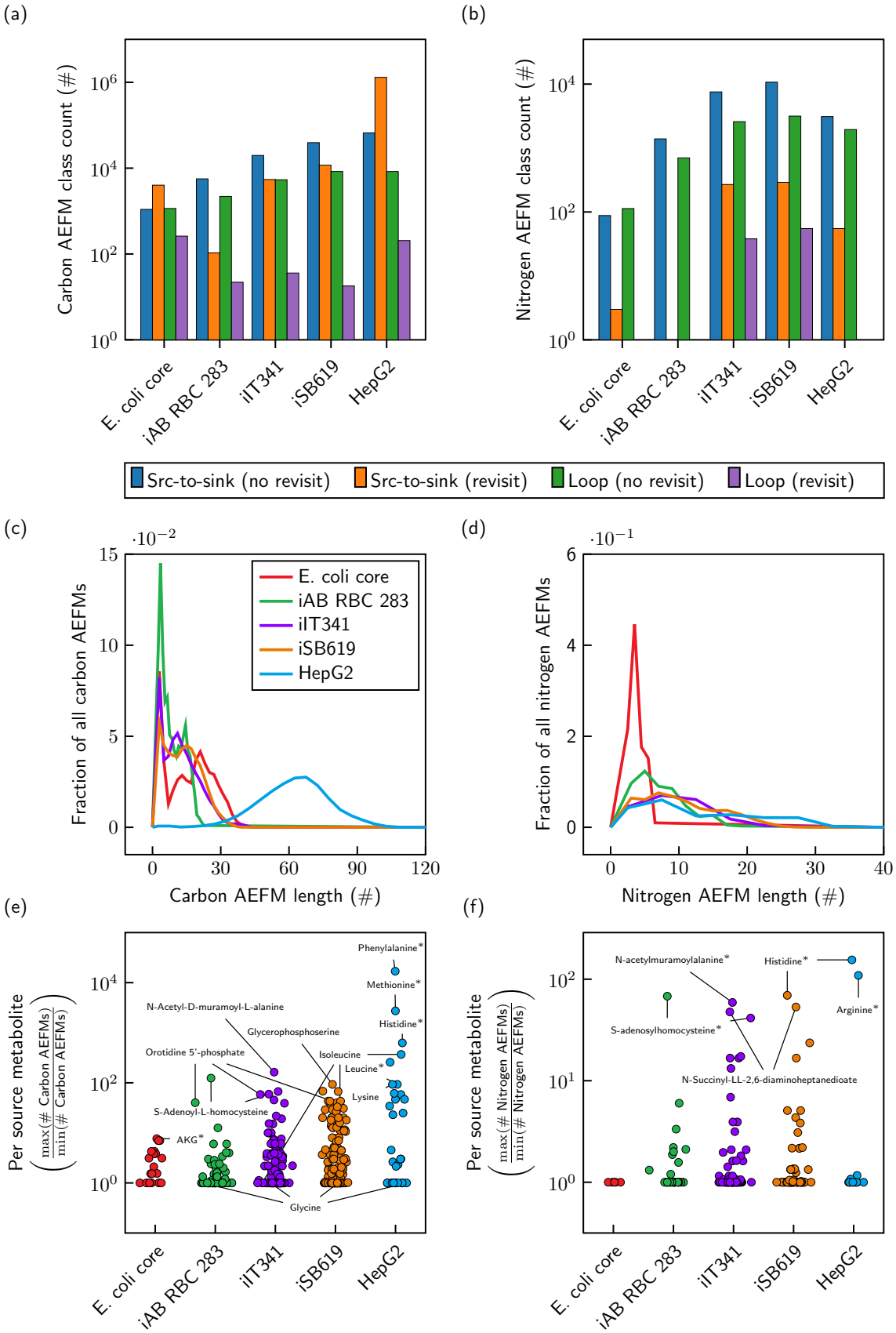

Figure 3: Structural analysis of AEFMs in five genome-scale metabolic models. (a-b) Categories of carbon and nitrogen EFMs. (c-d) Distribution carbon and nitrogen AEFM lengths. (e-f) Ratio of the greatest and smallest number of AEFMs between carbons and nitrogens within the same source metabolite. Representative metabolites from each dataset are labelled; labels with asterisks indicate that metabolite is also present in other networks.

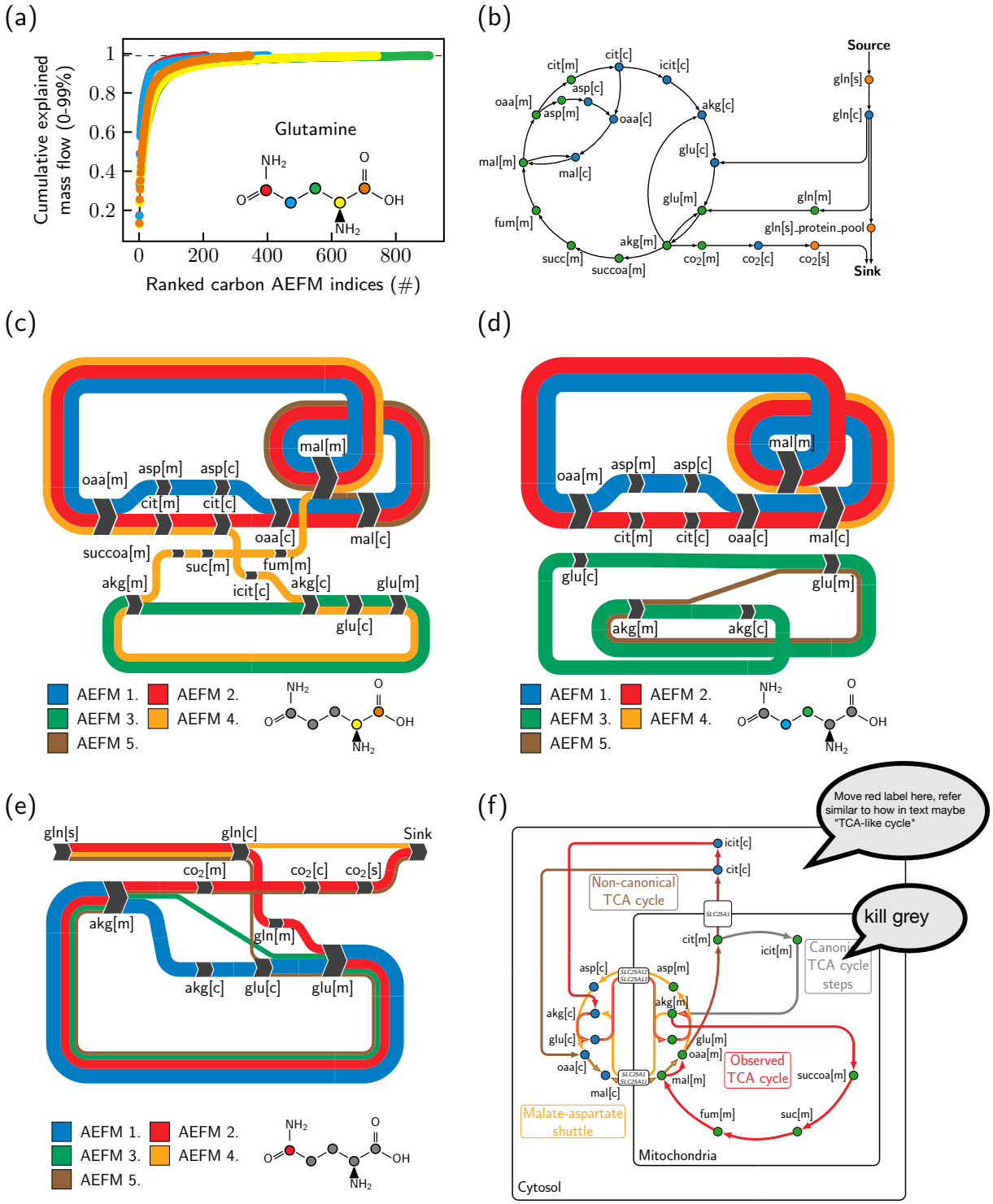

Figure 4: Carbon AEFM weight analysis of the HepG2 metabolic flux network. (a) A small number of highly active AEFMs explain the majority of glutamine flow. (b) Subnetwork of the top five AEFMs across each glutamine carbon collapsed onto metabolites. (c-e) Sankey diagrams of the top five glutamine carbon AEFMs. (f) Schematic mapping AEFMs onto traditionally defined metabolic pathways/subsystems. All metabolite names are labelled following the BiGG database nomenclature.